

Learning objectives

In this lesson students will learn to:

- add together two positive 8-bit binary patterns
- define what is meant by the term 'overflow error'
- describe the effects of an overflow error.

For more detail on this topic, and additional student activities, refer to Topic 2 of the student book.

Adding in binary

One of the most common operations that you will need to carry out is adding binary numbers together.

The method you will use is the same method you may have been taught to add denary numbers when you were younger: column addition.

Golden rules

There are 4 rules to remember. If you always apply them, you'll get your binary additions right every time.

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 1 = 0 \text{ carry } 1$$

$$1 + 1 + 1 = 1 \text{ carry } 1$$

Addition

1		0	0	0	1	
4	+	0	1	0	0	+
5		0	1	0	1	

1		0	0	0	1	
1	+	0	0	0	1	+
2				1		
		0	0	1	0	

In column 1, $1 + 1 = 2$; so write a 0 down and carry 1 over into the next column on the left.

3		0	0	1	1	
7	+	0	1	1	1	+
10		1	1	1		
		1	0	1	0	

In column 2, $1 + 1 + 1 = 3$ lots of 2 = 1 lot of 4 plus 1 lot of 2; so write a 1 down and carry 1 over into the next column on the left.

Overflow

Let's try one more together:

$$\begin{array}{rcccc}
 1 & 1 & 0 & 0 & & 0 & 1 & 1 & 1 \\
 1 & 1 & 1 & 1 & & 0 & 1 & 0 & 0 & + \\
 \hline
 1 & 0 & 1 & 1 & & 1 & 0 & 1 & 1 \\
 \hline
 1 & & & & & 1 & & & &
 \end{array}$$

Is this answer correct?

What's happened to the last carry digit?

Overflow

What you've just seen is called **overflow**.

It means the result of the calculation is too large for the space allocated (in this case 1 byte).

The way a computer handles this will depend on how it's setup.

- It could crash and report an overflow error.
- It could truncate the answer (simply leave off the extra one).
- It could wrap the number around back to zero.

Handling overflow in a manual calculation

Let's look again at our previous example:

$$\begin{array}{r}
 1\ 1\ 0\ 0\quad 0\ 1\ 1\ 1 \\
 1\ 1\ 1\ 1\quad 0\ 1\ 0\ 0\ + \\
 \hline
 1\ 0\ 1\ 1\quad 1\ 0\ 1\ 1 \\
 \hline
 1\qquad\qquad 1
 \end{array}$$

We know there is one final digit to represent, but because it has overflowed, we need to be clear that this is the case.

Our final answer should look something like this:

$$\begin{array}{r}
 1\ 1\ 0\ 0\quad 0\ 1\ 1\ 1 \\
 1\ 1\ 1\ 1\quad 0\ 1\ 0\ 0\ + \\
 \hline
 (1)\ 1\ 0\ 1\ 1\quad 1\ 0\ 1\ 1 \\
 \hline
 1\qquad\qquad 1
 \end{array}$$

In programming...

Python is a language that is flexible with storage space.

Integers are not limited to a specific number of bits and can expand to fit available memory.

In some applications, it's important to work out how much space is 'just enough', particularly when memory might be short.

Such applications include embedded systems where the hardware is basic.

Wrap up: you have learned how to...

- ✓ Add together two positive 8-bit binary integers.
 - Using the golden rules:
$$0 + 0 = 0$$
$$1 + 0 = 1$$
$$1 + 1 = 0 \text{ carry } 1$$
$$1 + 1 + 1 = 1 \text{ carry } 1$$
- ✓ Define what is meant by the term 'overflow error'.
 - When the result of a calculation is too large for the space allocated.
- ✓ Describe the effects of an overflow error.
 - An error could be reported, the value could be truncated, or the number could be wrapped around to 0.